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SVIT: Scaling up Visual Instruction Tuning

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Abstract

Thanks to the emerging of foundation models, the large language and vision models are integrated to acquire the multimodal ability of visual captioning, question answering, etc. Although existing multimodal models present impressive performance of visual understanding and reasoning, their limits are still largely under-explored due to the scarcity of high-quality instruction tuning data. To push the limits of multimodal capability, we Scale up Visual Instruction Tuning (SVIT) by constructing a dataset of 4.2 million visual instruction tuning data including 1.6M conversation question-answer (QA) pairs, 1.6M complex reasoning QA pairs, 1.0M referring QA pairs and 106K detailed image descriptions. Besides the volume, the proposed dataset is also featured by the high quality and rich diversity, which is generated by prompting GPT-4 with the abundant manual annotations of images. We also propose a new data recipe to select subset with better diversity and balance, which evokes model's superior capabilities. Extensive experiments verify that SVIT-v1.5, trained on the proposed dataset, outperforms state-of-the-art Multimodal Large Language Models on popular benchmarks. The data and code are publicly available at https://github.com/BAAI-DCAI/Visual-Instruction-Tuning.

1. Introduction

The great success of large language models (LLMs), e.g. BERT (Devlin et al., 2019), T5 (Raffel et al., 2020), GPT-2 (Radford et al., 2019), GPT-3 (Brown et al., 2020), have motivated the advancement of vision (Dosovitskiy et al., 2021; Liu et al., 2021; He et al., 2022) and multimodality (Radford et al., 2021; Alayrac et al., 2022; Zhu et al., 2023a; Liu et al., 2023b) in terms of architecture design and learning paradigm. Recently, GPT-4 (OpenAI, 2023) demonstrates impressive multimodal understanding and reasoning abilities, accepting image and text inputs. Inspired by GPT-4, Multimodal Large Language Models (MLLMs) bridging language and vision models have achieved remarkable progress in multiple visual understanding and reasoning tasks, e.g. visual captioning (Li et al., 2023c), dialogue (Alayrac et al., 2022) and question answering (Zhu et al., 2023a; Liu et al., 2023b).

Typically, the multimodal models are pre-trained on large multimodal datasets, e.g. LAION-2B (Schuhmann et al., 2022), CC-12M (Changpinyo et al., 2021), YFCC-100M (Thomee et al., 2016) and MMC4 (Zhu et al., 2023b), that contain millions to billions roughly-aligned image-text pairs from the web. Then, precise vision-language data pairs are used to finetune the models. Like the success of language instruction tuning, visual instruction tuning has become the key to the multimodal performance. However, due to the high construction cost, existing visual instruction datasets are still in small scale and less informative. Several works convert the image captioning and VQA datasets (Lin et al., 2014; Antol et al., 2015; Hudson & Manning, 2019; Goyal et al., 2017) into instruction tuning data by manually adding a few instructions (Dai et al., 2023). However, these captions and questions/answers are usually short and focus on visual perception and simple questions, which may lead to ineffective model training (Gong et al., 2023). To generate more informative visual instruction data, GPTs are introduced. LLaVA (Liu et al., 2023b) contributes a large visual instruction dataset containing 158K data by prompting GPT-4 with five captions and a few object bounding boxes associated with images from COCO dataset (Lin et al., 2014). Meanwhile, MiniGPT-4 (Zhu et al., 2023a) creates 3,500 image-text pairs by refining model's output using ChatGPT. The language-only GPT models have difficulty in precisely imagining the whole picture from the limited input. Thus, the generated instruction tuning data lacks diversity and complexity.

To push the limits of large multimodal models, we Scale up Visual Instruction Tuning (SVIT) and propose a large-scale dataset with 4.2 million informative instruction tuning data, including 1.6M conversation QA pairs, 1.6M complex reasoning QA pairs, 1.0M referring QA pairs and 106K detailed descriptions. Table 1 shows that SVIT is 20×

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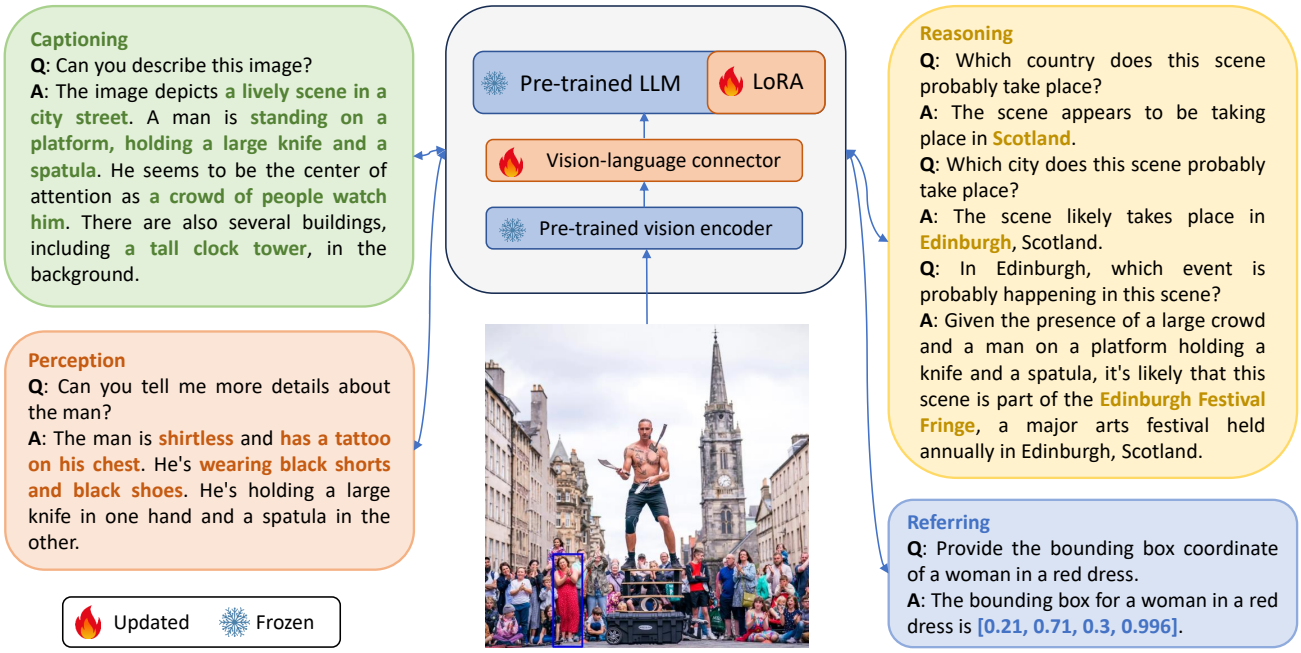

Figure 1. SVIT-v1.5 (LoRA) model architecture and abilities.

larger than LLaVA dataset. To enrich the diversity and informativeness of instruction tuning data, we construct SVIT based on Visual Genome (Krishna et al., 2017) which has abundant manual annotations and GPT-4 which has the best multimodal capability. We prompt the language-only GPT-4 ChatBot with image-level descriptions, detailed region descriptions and object bounding boxes. We further study the data efficiency and propose a new data recipe that outputs subset with better diversity and balance. Then, a more powerful model, SVIT-v1.5, is trained on the proposed dataset, as illustrated in Figure 1. Extensive experiments verify that our model reveals impressive ability in visual perception and reasoning, and achieves noticeable performance improvements over the state of the art.

We summarize the main contributions of this paper:

1. We present 4.2M high-quality instruction data of 1.6M conversation QA pairs, 1.6M complex reasoning QA pairs, 1.0M referring QA pairs and 106K detailed image descriptions.
2. We propose a new data recipe that selects an informative subset of diverse and balanced training data to better match the downstream tasks.
3. We scale up visual instruction tuning and contribute a better model – SVIT-v1.5 that outperforms state-of-the-art MLLMs including LLaVA-v1.5, Qwen-VL-Chat and InstructBLIP on popular benchmarks.

2. Related Work

2.1. Multimodal Models

Existing multimodal solutions can be roughly split into two categories: 1) multimodal systems, e.g. Visual ChatGPT (Wu et al., 2023), X-Decoder (Zou et al., 2023) and InternGPT (Liu et al., 2023d), in which multiple language and vision models are coordinated by a LLM manager/controller to deal with different tasks, 2) end-to-end differentiable multimodal models, e.g. Flamingo (Alayrac et al., 2022), BLIP-2 (Li et al., 2023c), Kosmos (Huang et al., 2023; Peng et al., 2023), MiniGPT-4 (Zhu et al., 2023a), LLaVA (Liu et al., 2023b), InstructBLIP (Dai et al., 2023) which input both vision and language tokens into LLM. In this paper, we focus on the end-to-end differentiable multimodal models, which are lightweight and concise for research.

The end-to-end multimodal models contain pre-trained vision and language models and a learnable module to fuse both. Flamingo (Alayrac et al., 2022) learns gated cross-attention layers to condition the frozen LLM on visual tokens, demonstrating excellent in-context few-shot learning performance. Li et al. (2023c) design Q-Former to bridge the image encoder and LLM in a two-stage training strategy, which shows emerging capability of zero-shot instructed image-to-text generation. By leveraging advanced LLMs, i.e. LLaMA (Touvron et al., 2023) and Vicuna (Chiang et al., 2023), multimodal models LLaVA (Liu et al., 2023b) and MiniGPT-4 (Zhu et al., 2023a) are built by transforming

Table 1. Comparing SVIT to similar vision-language instruction datasets generated by GPT. *LLaVAR collects 422K noisy instruction-following data using OCR results and 16K high-quality data using GPT-4.

Dataset	#Image	#Object BBox	#Region Description	#Image Caption	#Instruction Question	#Response Answer	GPT
MiniGPT-4	3.5K	-	-	-	4	3.5K	GPT-3.5
LLaVAR*	16K	-	-	-	16K	16K	GPT-4
LLaVA	81.5K	600K	-	404.7K	158K	158K	GPT-4
SVIT	108.1K	3.8M	5.4M	257.6K	4.2M	4.2M	GPT-4

visual tokens to language tokens with only one linear layer, while InstructBLIP (Dai et al., 2023) learns a Q-Former to bridge vison and language models.

2.2. Multimodal Instruction Tuning

The success of multimodal models, e.g. LLaVA, MiniGPT-4 and InstructBLIP, relies on the high-quality image-text data for finetuning models, which is named visual instruction tuning in Liu et al. (2023b). Previous work (Gong et al., 2023) finds that simply constructing training set based on existing VQA datasets (Goyal et al., 2017; Hudson & Manning, 2019) with short answers will degrade the model performance. To boost the performance, Zhu et al. (2023a) collect 3,500 high-quality image-text pairs by refining their model's outputs using ChatGPT. More natural and reliable responses are produced by finetuning the model on the refined data. Liu et al. (2023b) for the first time systematically construct a large visual instruction tuning dataset – LLaVA-Instruct-150K. They prompt GPT-4 to generate questions and answers by feeding it image-level captions and object bounding boxes of each image from COCO dataset (Lin et al., 2014). To better understand text-rich images, Zhang et al. (2023c) present LLaVAR that collects 422K noisy instruction-following data using OCR results and 16K high-quality data using GPT-4. Dai et al. (2023) collect 26 public datasets including LLaVA-Instruct-150K to construct visual instruction tuning data. However, most of these public datasets contain short questions and answers that focus on visual perception. Li et al. (2023d) build M^3IT by converting 40 datasets into a unified vision-to-text schema. They utilize ChatGPT to paraphrase the short answers in original VQA datasets. Beyond above works, we prompt the powerful GPT-4 with rich annotations of image-level captions, region-level descriptions and object bounding boxes that are from Visual Genome (Krishna et al., 2017) and COCO dataset (Lin et al., 2014). The generated 4.2M visual instruction data cover diverse tasks of visual perception, reasoning and planing.

There are also some works that contribute multimodal instruction data of videos (Li et al., 2023b), RGB-D images (Li et al., 2023b), speech (Zhang et al., 2023a), audio (Zhang et al., 2023b), etc. For instance, EgoCOT (Mu et al., 2023) prompts ChatGPT with video captions to generate instructions and responses of detailed embodied planning. MIMIC-IT (Li et al., 2023b) collects visual data from multiple datasets, and prompts ChatGPT to generate instruction-response pairs. Most of its data are constructed based on the egocentric videos from E4D dataset (Grauman et al., 2022).

3. Dataset Construction

3.1. Source Data

We build SVIT based on Visual Genome (Krishna et al., 2017) dataset that comprises 108,077 images with dense annotations within each image, including region descriptions, objects, attributes, relationships, etc. Since Visual Genome is partially sourced from COCO dataset (Lin et al., 2014), we also collect captions for images from COCO dataset. Generally, each image in COCO dataset has 5 captions, focusing on the high-level appearance. As an image usually contains rich objects and regions that cannot be completely described in a general caption, Visual Genome serves as a valuable source, offering abundant annotations of the visual details. On average, Visual Genome provides 42 human-generated region descriptions and 21 objects per image, with each region and object located by a bounding box. Leveraging these annotations, we are able to gather thorough and detailed descriptions for all images, which are made up of three key components: (1) the 257,633 captions from COCO dataset; (2) the 3,802,374 object names and their corresponding bounding boxes from Visual Genome; (3) the 5,406,592 region descriptions and their corresponding bounding boxes from Visual Genome.

3.2. Instruction Data Generation

Inspired by LLaVA (Liu et al., 2023b), we design four tasks and prompt the language-only GPT-4 ChatBot to generate the questions and answers accordingly. The prompts are summarized in Figure 7 and Figure 8 in the Appendix. Since GPT-4 demonstrates excellent performance even with zero-shot learning, we do not provide any examples for GPT-4 in order to encourage the innovation and diversity of the generated contents.

Input

(1) captions:
Street workers in orange jackets outside working with a traffic signal.
The workers are trying to install a broken signal.
…
(2) objects:
man: [0.588, 0.387, 0.692, 0.832]
traffic light: [0.16, 0.341, 0.34, 0.571]
…
(3) regions:
four men installing a traffic light: [0.032, 0.197, 0.992, 0.952]
Dirt on the street: [0.13, 0.739, 0.63, 0.856]
A traffic light leaning on the truck: [0.146, 0.355, 0.984, 0.808]
…

Output

Conversation:
Topic: Traffic Light Installation
User: How many workers are present in the image?
GPT: There are four workers actively involved in the installation of the traffic light in the image.
User: Can you describe the clothing of the workers?
GPT: The workers are wearing orange jackets, or vests with shirts underneath. The jackets and vests are safety attire, which is typical for street workers to make them more visible to motorists.
User: Are there any safety measures visible in the image?
GPT: Yes, as a safety measure, there are orange cones with white stripes placed on the street near the working area to alert motorists and pedestrians of the ongoing construction.
…
Complex reasoning:
Question: What precautions are the workers taking to ensure their safety?
Answer: The workers are wearing orange jackets, which increase their visibility. Additionally, they've placed orange cones around the area where they are working.
Question: What could be a recommendation to improve safety at this work site?
Answer: Adding more clear signage to alert oncoming traffic of the construction work ahead could be one recommendation to improve safety.
…
Referring QAs:
Question: Why might the <st>car<ed> [0.386, 0.435, 0.996, 0.715] be stopped?
Answer: The car might be stopped due to the ongoing construction and the presence of workers on the street fixing the traffic light.
Question: What suggestion would you give to improve safety for the workers?
Answer: It would be good to place more <st>cones<ed> [0.354, 0.573, 0.774, 0.875] around the working area and possibly use barriers to ensure vehicles are aware and kept at a distance from the ongoing construction.
…
Detail description:
In the image, I see a scene where street workers in orange jackets are outside, working with a traffic signal. … There is a man wearing an orange shirt, and he appears to be the main focus of the image. He is accompanied by several other men who are also wearing orange shirts. They are engaged in various activities, such as fixing a street light in the middle of the street. The workers are standing on a street with dirt and orange cones scattered around. In the background, there is a white car and some windows can be seen on it. … Overall, this image captures a group of workers actively engaged in road construction and traffic signal maintenance.

Figure 2. The example input to GPT-4 and the responses for three tasks. Note that the image is only shown here for reference and not provided to GPT-4. The colored phrases in referring QAs correspond with bounding boxes of that color in the image.

- **Conversation.** We prompt GPT-4 to design 3 conversations between a person and GPT-4 talking about the image. Each conversation should include 5 question and answer pairs (QAs). The content of the conversation should be logically connected. GPT-4 thinks about the topic first and then generates the conversation according to the topic. The topics can be about the visual perception, reasoning, event planning, etc.

- **Complex reasoning.** 15 complex reasoning QAs about each image are generated using GPT-4. The questions can be asking why things happen that way, suggestions to the people in the image, etc. When providing the answer to a complex question, we prompt GPT-4 to think step by step and include reasoning details in the answer.

- **Referring QAs.** We prompt GPT-4 to create 10 question and answer pairs of specific regions in the image. When referring to any object in the question or answer, always wrap it with prefix "<st>", suffix "<ed>" and attach its normalized bounding box after it, in the format of "<st>object<ed> [x1, y1, x2, y2]". If multiple objects are referred to, attach all the corresponding bounding boxes after them, e.g., "<st>objects<ed> [x1, y1, x2, y2], [x1, y1, x2, y2]".

- **Detail description.** We use GPT-4 to describe the image in detail. The description may include appearances, actions, the number of objects, object positions, background details, etc.

Figure 2 illustrates an example input and the GPT-4 output for each task. For rich diversity, we further randomly sample an instruction for detail description task, e.g., "can you describe the image in detail". The complete list of the alternative instructions can be found in Figure 9 in the Appendix.

3.3. Postprocessing

While most of the GPT-4 generated question-answer pairs are of high quality, some answers occasionally contain unneeded contents. For example, some answers may tell that the information is based on the given "captions" and "descriptions". To remove the unneeded content, we find them based on relative words and use GPT-4 to regenerate the responses. In addition, the number of generated conversations or QA pairs may be fewer than the requirement. We also remove them and generate new responses. We use the same procedure to filter the regenerated content until it is satisfying.

3.4. Statistics and Analysis

Statistics. Employing the two-pass procedure, we obtain an extensive collection of data, including 1,565,797 conversation QAs, 1,556,902 complex reasoning QAs, 1,011,338 referring QAs and 106,274 detailed image descriptions. The averaging question and answer lengths are 9.6 and 27.9 words in *conversation* subset, 12.6 and 26.6 words in *complex reasoning* subset and 11.3 and 20.6 words in *referring QAs* subset, respectively. In contrast, the mean length is 5.7 words per question and 1.8 words per answer in the original Visual Genome. The detailed descriptions in our dataset have 361.5 words on average, while the length of COCO dataset image captions is 11.3. Therefore, the corpus provided by our SVIT is of higher quality.

Distribution. We analyze the distribution of question types in *conversation*, *complex reasoning* and *referring QAs* tasks by visualizing the distribution of first three words in Figure 3. We can see that "what" questions are the largest category, in *conversation* (38%), *complex reasoning* (55%) and *referring QAs* (41%). In the case of *conversation*, question types are diverse, including simple yes-no questions, questions on object details, conditions and functions, etc. Regarding *complex reasoning*, since we explicitly prompt GPT-4 to generate questions that need complex reasoning to answer, we collect a larger proportion of complex questions that commence with "why" (9%) and "how" (11%). Furthermore, most questions starting with "how" are simple object counting questions, i.e. "how many", in existing visual question answering datasets such as Visual Genome (Krishna et al., 2017) and VQA (Goyal et al., 2017), while in SVIT, only 11% of questions starting with "how" are the "how many" questions. For *referring QAs*, there are various types of questions, including those about object positions that start with "where", about object existence that start with "is/are there any", about suggestions and planning that start with "what suggestion" and about reasoning that start with "why". To better distinguish objects in the same image, there is also a notable proportion of questions that starts with "which".

Correctness. To assess the correctness of the generated content, we conduct a manual examination on randomly selected 20 images and corresponding data. In general, around 5% of the questions in the dataset can be provided with a more accurate or satisfying answer. The identified problems can be categorized into three types.

- Errors in original annotations. We construct the visual instruction data based on the manual annotations from Visual Genome and COCO dataset, which may contain errors in their original annotations. For example, in the image depicted in Figure 4(a), one caption from

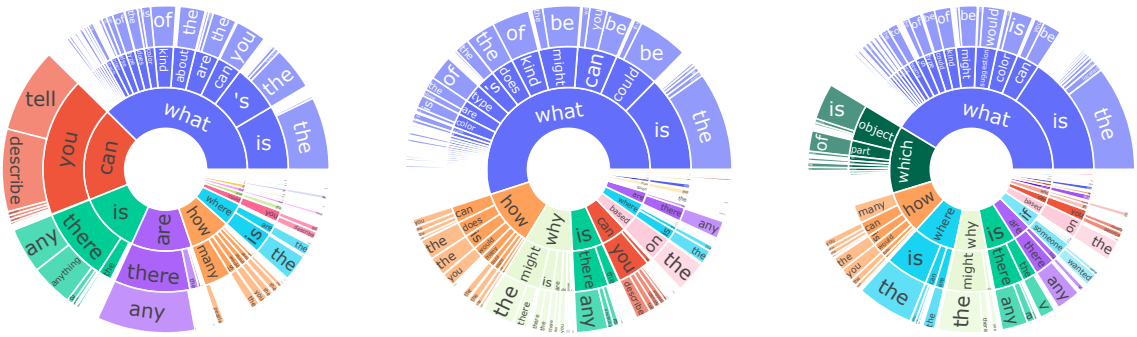

Figure 3. The distribution of question types in *conversations* (left), *complex reasoning* (middle), *referring QAs* (right) by the first three words. The angle of each sector represents the proportion of each category.

(a) Wrong caption in COCO dataset: "Three men and one older woman stand near a man who is looking in the mirror with the collar of his white shirt up."

(b) Wrong object name in Visual Genome: "teddy bear".

(c) The answer discusses how the condition of the boat's paint would reflect the maintenance instead of answering it directly.

(d) The generated answer misunderstands the position of the telephone.

Figure 4. Problematic examples in generated answers.

COCO dataset incorrectly states, "Three men and one older woman stand near a man who is looking in the mirror with the collar of his white shirt up." Actually, there are only two men and one woman standing near the man looking at the mirror. Similarly, in Figure 4(b), the object is labeled as a "little bunny" in the region description, but wrongly referred to a "teddy bear" in the object name in Visual Genome's annotation.

- Correct but not precisely answer the question. As illustrated in Figure 4(c), when being asked, "What can be inferred about the maintenance of the boat from the condition of the paint?", the answer states, "The condition of the boat's paint could reflect the level of maintenance, if it's faded or peeling, it may suggest the boat hasn't been maintained well, whereas bright and fresh paint may indicate regular upkeep." Although the answer is correct, it fails to address the question precisely.

- Incorrect answers. In Figure 4(d), the generated image description mentions, "Nearby, there's a round center table cluttered with assorted magazines and books, creating a lived-in feel. The table also hosts a yellow rotary telephone, a vintage relic of bygone days." In reality, there are two tables in the image and the telephone is placed on a different table in the bottom left corner, though it needs careful observation.

4. Method

4.1. Model Architecture

We employ the open-source Multimodal Large Language Model - LLaVA (Liu et al., 2023b;a), which consists of a vision encoder $\psi_{\mathrm{V}}(\cdot, \boldsymbol{\theta}_{\mathrm{V}})$, a large language model $\psi_{\mathrm{L}}(\cdot, \boldsymbol{\theta}_{\mathrm{L}})$ and a vision-language connector $\psi_{\mathrm{C}}(\cdot, \boldsymbol{\theta}_{\mathrm{C}})$. We illustrate the model in Figure 1. Provided with the input image $\boldsymbol{x}_{\mathrm{V}}$ and instruction $\boldsymbol{x}_{\mathrm{I}}$, the vision encoder is utilized to extract the image features $\boldsymbol{f} = \psi_{\mathrm{V}}(\boldsymbol{x}_{\mathrm{V}}, \boldsymbol{\theta}_{\mathrm{V}})$. Then a vision-language connector is applied to convert the image features to the language embedding tokens $\psi_{\mathrm{C}}(\boldsymbol{f}, \boldsymbol{\theta}_{\mathrm{C}})$. After that, the vision and language tokens are combined and fed into the LLM to generate the response:

$$\tilde{\boldsymbol{x}_{\mathrm{R}}} = \psi_{\mathrm{L}}([\psi_{\mathrm{C}}(\boldsymbol{f}, \boldsymbol{\theta}_{\mathrm{C}}), \boldsymbol{x}_{\mathrm{I}}], \boldsymbol{\theta}_{\mathrm{L}}) \tag{1}$$

The training procedure contains two stages, including the pre-training on image-text pairs and fine-tuning on visual instruction data. In the pre-training stage, the vision-language connector parameters are updated using image-text pairs, while the weights of vision encoder and LLM remain frozen. In the fine-tuning stage, we implement full-parameter tuning or Low-rank Adaption (LoRA) tuning (Hu et al., 2022). Without ambiguity, $\boldsymbol{\theta}_{\mathrm{L}}$ denotes the LLM parameters in full training setting and the learnable LoRA parameters in LoRA training setting. Then, the connector and learnable LLM parameters are updated using visual instruction data:

$$\boldsymbol{\theta}_{\mathrm{C}}^*,\ \boldsymbol{\theta}_{\mathrm{L}}^* = \underset{\boldsymbol{\theta}_{\mathrm{C}},\ \boldsymbol{\theta}_{\mathrm{L}}}{\arg\min} -\sum_{i=1}^{N}\sum_{j=1}^{L} \log p(\tilde{\boldsymbol{x}_{\mathrm{R}}}_i^j | \boldsymbol{x}_{\mathrm{V}i}, \boldsymbol{x}_{\mathrm{I}i}, \boldsymbol{x}_{\mathrm{R}}{}_i^{<j}), \tag{2}$$

where N and L denote the training sample size and the length of each response.

4.2. Coreset Selection Algorithm

The popular benchmarks evaluate different abilities of Multimodal Large Language Models (MLLM), which require specific recipe of training data to evoke the pre-trained model. Thus, we design a new data recipe, i.e. a coreset selection algorithm, to better adapt those benchmarks and achieve balance between performance and training efficiency.

Diversity. We construct a set of key concepts that match the popular benchmarks, namely, MME (Fu et al., 2023) and MMBench (Liu et al., 2023c). Specifically, we design several high-level concepts and then use GPT-4 to generate dozens of key words about each concept. Then, we filter out those key words that have low frequency in SVIT dataset. The concept set is illustrated in Table 4 in the Appendix. We measure the informativeness of each training sample by its overlap with concept set, and select the most informative ones.

Balance. "Yes" or "No" questions are used to evaluate models in MME benchmark. However, the proportion of the two choices in GPT-4 generated data is extremely unbalanced, which makes the tuned model has tendency to respond "Yes". We adjust the proportion by re-sampling. We empirically study the relation between "Yes:No" proportion and model performance in Section 5.2.

With the above two operations, we obtain the coreset **SVIT-core-150K** of 157,712 samples, which has the same size as LLaVA-Instruct-150K. We also produce **SVIT-mix-665K** by replacing LLaVA-Instruct-150K in LLaVA-v1.5-mix-665K (Liu et al., 2023a) with SVIT-core-150K.

5. Experiments

Firstly, we compare our model to the state-of-the-art MLLMs in Section 5.1. In this sub-section, we tune the advanced LLaVA-v1.5-13B (Liu et al., 2023a) on the constructed SVIT-mix-665K dataset. Secondly, we implement ablation study and provide more detailed evaluations in Section 5.2. We tune the LLaVA-v1.0 (LLaVA-LLaMA-2-7B-Chat) (Liu et al., 2023b) with various data recipes for efficiency. Lastly, qualitative evaluation is provided in Section 5.3.

Table 2. Comparison to state-of-the-art MLLMs on 11 benchmarks. Our models outperform LLaVA-v1.5 and others in most of the settings. We evaluate these models on benchmarks: VQA-v2 (Goyal et al., 2017) test-dev split, GQA (Hudson & Manning, 2019) test-dev-balanced split, VisWiz (Gurari et al., 2018) test-dev split, SQA^I: ScienceQA-IMG (Lu et al., 2022) test split, VQA^T: TextVQA (Singh et al., 2019) validation split, MME^P: MME perception (Fu et al., 2023), MME^C: MME cognition (Fu et al., 2023), MMB: MMBench (Liu et al., 2023c) test split, MMB^{CN}: MMBench-Chinese (Liu et al., 2023c) test split, SEED: SEED-Bench (Li et al., 2023a), and MMMU (Yue et al., 2023) test split. We mark the best performance **bold** and the runner-up underlined. *The training images of the datasets are observed during training. §We evaluate the officially released checkpoint by ourselves.

Method	LLM	VQA^{v2}	GQA	VisWiz	SQA^I	VQA^T	MME^P	MME^C	MMB	MMB^{CN}	SEED	MMMU
BLIP-2	Vicuna-13B	–	41.0	19.6	61.0	42.5	–	–	–	–	–	–
BLIP-2	Flan-T5-XXL	65.0	44.6	29.4	64.5	44.1	1293.8	290.0	–	–	–	34.0
InstructBLIP	Vicuna-7B	–	49.2	34.5	60.5	50.1	–	–	33.9	23.9	53.4	–
InstructBLIP	Vicuna-13B	–	49.5	33.4	63.1	50.7	–	–	–	–	–	–
InstructBLIP	Flan-T5-XXL	–	47.9	30.9	70.6	46.6	–	–	–	–	–	33.8
Shikra-7B	Vicuna-7B	–	–	–	–	–	–	–	60.2	–	–	–
Shikra-13B	Vicuna-13B	77.4*	–	–	–	–	–	–	–	–	–	–
IDEFICS-9B	LLaMA-7B	50.9	–	35.5	44.2	25.9	–	–	45.3	25.2	–	–
IDEFICS-80B	LLaMA-65B	60.0	–	36.0	68.9	30.9	–	–	54.6	38.1	–	–
Qwen-VL	Qwen-7B	79.5*	59.3*	35.2	67.1	**63.8***	–	–	32.2	7.8	56.3	–
Qwen-VL-Chat	Qwen-7B	78.2*	57.5*	38.9	68.2	61.5*	1487.6	360.7	61.8	56.3	58.2	32.9
mPLUG-Owl2	LLaMA2-7B	79.4*	56.1*	54.5	68.7	58.2	1450.2	313.2	66.0	60.3	57.8	32.1
LLaVA-v1.5 (LoRA)	Vicuna-13B	80.0*	63.3*	**58.9**	71.2	60.2	1541.7	300.4§	68.4§	62.4§	61.3	33.2§
LLaVA-v1.5 (Full)	Vicuna-13B	80.0*	63.3*	53.6	**71.6**	61.3	1531.3	295.4	67.8	**63.3**	61.6	33.6
SVIT-v1.5 (LoRA)	Vicuna-13B	80.1*	63.4*	56.7	69.9	61.1	1560.3	**364.3**	68.3	63.2	61.8	**34.1**
SVIT-v1.5 (Full)	Vicuna-13B	**80.3***	**64.1***	56.4	70.0	60.8	**1565.8**	323.2	**69.1**	63.1	**61.9**	33.3

5.1. Comparison to the State of the Art

We adopt LLaVA-v1.5-13B (Liu et al., 2023a) architecture and pre-training weights, and then tune it on the constructed SVIT-mix-665K, which is named **SVIT-v1.5**. Specifically, we replace LLaVA-v1.5-mix-665K with our SVIT-mix-665K in the visual instruction tuning stage. The rest of model training protocol is kept unchanged for fair comparison. Visual instruction tuning takes about 21 hours for both full-parameter tuning and LoRA tuning on 8 NVIDIA Tesla A100 GPUs, each with 80GB memory, with DeepSpeed ZeRO Stage 3. We compare SVIT-v1.5 to state-of-the-art MLLMs: BLIP-2 (Li et al., 2023c), InstructBLIP (Dai et al., 2023), Shikra (Chen et al., 2023), IDEFICS (IDEFICS, 2023), Qwen-VL(-Chat) (Bai et al., 2023), mPLUG-Owl2 (Ye et al., 2023) and LLaVA-v1.5 (Liu et al., 2023a). We evaluate these models on popular benchmarks: VQA-v2 (Goyal et al., 2017), GQA (Hudson & Manning, 2019), VisWiz (Gurari et al., 2018), ScienceQA-IMG (Lu et al., 2022), TextVQA (Singh et al., 2019), MME perception (Fu et al., 2023), MME cognition (Fu et al., 2023), MMBench (Liu et al., 2023c), MMBench-Chinese (Liu et al., 2023c), SEED-Bench (Li et al., 2023a) and MMMU (Yue et al., 2023).

As shown in Table 2, our SVIT-v1.5 outperforms LLaVA-v1.5 and other models in most settings. Especially, in the most popular benchmark - MME, SVIT-v1.5 (Full) achieves 1565.8 score in MME perception and overwhelms LLaVA-v1.5 (Full) by 34.5 score. In the efficient LoRA training setting, SVIT-v1.5 (LoRA) exceeds LLaVA-v1.5 (LoRA) by 63.9 score, namely, 364.3 v.s. 300.4, in MME cognition. The improvements verify the better training effects of SVIT data, since the same data amount and base model are used.

5.2. Ablation Study

We further study the data quality, diversity strategy, balance strategy and scaling-up effects. 10% of the images are randomly sampled from SVIT as the held-out testing set for evaluation. The training split is denoted as **SVIT-train**. Note that, for saving the training cost, we implement ablation study with the LLaVA-v1.0 model and evaluate on MME benchmark. We denote the LLaVA-v1.0 model trained on SVIT data as SVIT-v1.0.

Data Quality. LLaVA-v1.0 employs LLaVA-Instruct-80K as the visual instruction tuning data. To demonstrate the quality of SVIT, we construct a subset of SVIT-train at same scale of LLaVA-Instruct-80K and fine-tune LLaVA-v1.0 by replacing LLaVA-Instruct-80K with the SVIT subset. Without loss of generality, the subset is constructed by randomly sampling 20K data from *conversation*, *complex reasoning*, *referring QAs* and *detail description*, leading to a subset of 80K data in total, denoted as **SVIT-80K**. We adopt the same training protocol and hyper-parameters as LLaVA-v1.0. The

Table 3. Evaluating models fine-tuned on LLaVA-Instruct-80K, SVIT-80K (random selection), SVIT-80K-D (enhancing diversity), SVIT-80K-B (with "Yes/No" balancing) and SVIT-train (SVIT train split) on MME benchmark. Note that the base model is LLaVA-v1.0 (Liu et al., 2023b). For LLaVA-Instruct-80K, we evaluate the officially released checkpoint by ourselves.

Task	Sub-task	LLaVA-Instruct-80K	SVIT-80K	SVIT-80K-D	SVIT-80K-B	SVIT-train
Overall	Total	1147.70	1241.84	1262.15	1329.77	1399.66
Perception	Total	906.63	1005.41	1017.15	1035.13	1166.45
	Existence	90.00	90.00	95.00	120.00	185.00
	Count	55.00	115.00	110.00	118.33	131.67
	Position	56.67	53.33	58.33	58.33	56.67
	Color	50.00	50.00	55.00	58.33	100.00
	Posters	116.33	143.20	146.26	133.67	134.01
	Celebrity	85.88	75.88	77.06	84.71	77.35
	Scene	152.75	161.25	159.50	153.75	153.25
	Landmark	130.75	148.25	151.00	137.75	144.50
	Artwork	96.75	111.00	107.50	105.25	104.00
	OCR	72.50	57.50	57.50	65.00	80.00
Cognition	Total	241.07	236.43	245.00	294.64	233.21
	Commonsense reasoning	83.57	86.43	80.00	87.14	80.71
	Numerical calculation	45.00	57.50	55.00	57.50	47.50
	Text translation	57.50	50.00	65.00	97.50	50.00
	Code reasoning	55.00	42.50	45.00	52.50	55.50

training takes less than 1 hour on 8 NVIDIA Tesla A100 GPUs, each with 40GB memory, with DeepSpeed ZeRO Stage 3.

The evaluation results on MME benchmark are shown in Table 3. The model fine-tuned on SVIT-80K achieves higher performance (+8.2%) than the model fine-tined by LLaVA-Instruct-80K. Specially, the model fine-tuned on SVIT-80K outperforms on "count" (+109.1%), "posters"(+23.1%), "scene (+5.6%)", "landmark (+13.4%)", "artwork (+14.7%)" in perception tasks, as well as "commense reasoning"(+3.4%), "numerical calculation" (+27.8%) in cognition tasks. The high performance of SVIT on those tasks can be due to the fact that SVIT dataset is constructed with more detailed manual annotations of the images, and the prompts for GPT-4 to generate QAs are carefully designed to cover a wide range of tasks, evoking the model to understand the images more accurately and comprehensively.

Figure 5. The relation between "Yes:No" proportion in training data and model performance.

Diversity Strategy. We produce an 80K subset selected with diversity strategy and compare it to the randomly selected SVIT-80K. We first remove half less-informative samples of SVIT-train based on the measured informativeness of each sample. Then we randomly sample 20K data for each category in SVIT, leading to the 80K subset – **SVIT-80K-D**. As showed in Table 3, its performance has an improvement of 20.3 score over SVIT-80K, which verifies the effectiveness of the diversity strategy.

Balance Strategy. The MME benchmark consists of 2,374 "Yes" or "No" answers with the proportion $1:1$. However, the randomly selected SVIT-80K dataset contains 7.5% "Yes" or "No" QA pairs with the proportion $Y:N=20$. We analyze the relation between the "Yes:No" proportion and model performance by adjusting the proportion. As shown in Figure 5, the model trained on randomly sampled SVIT-80K with $Y:N=20$ responds 1,393 "Yes" while 981 "No" on MME questions. We adjust the "Yes:No" proportion in training data by randomly dropping some

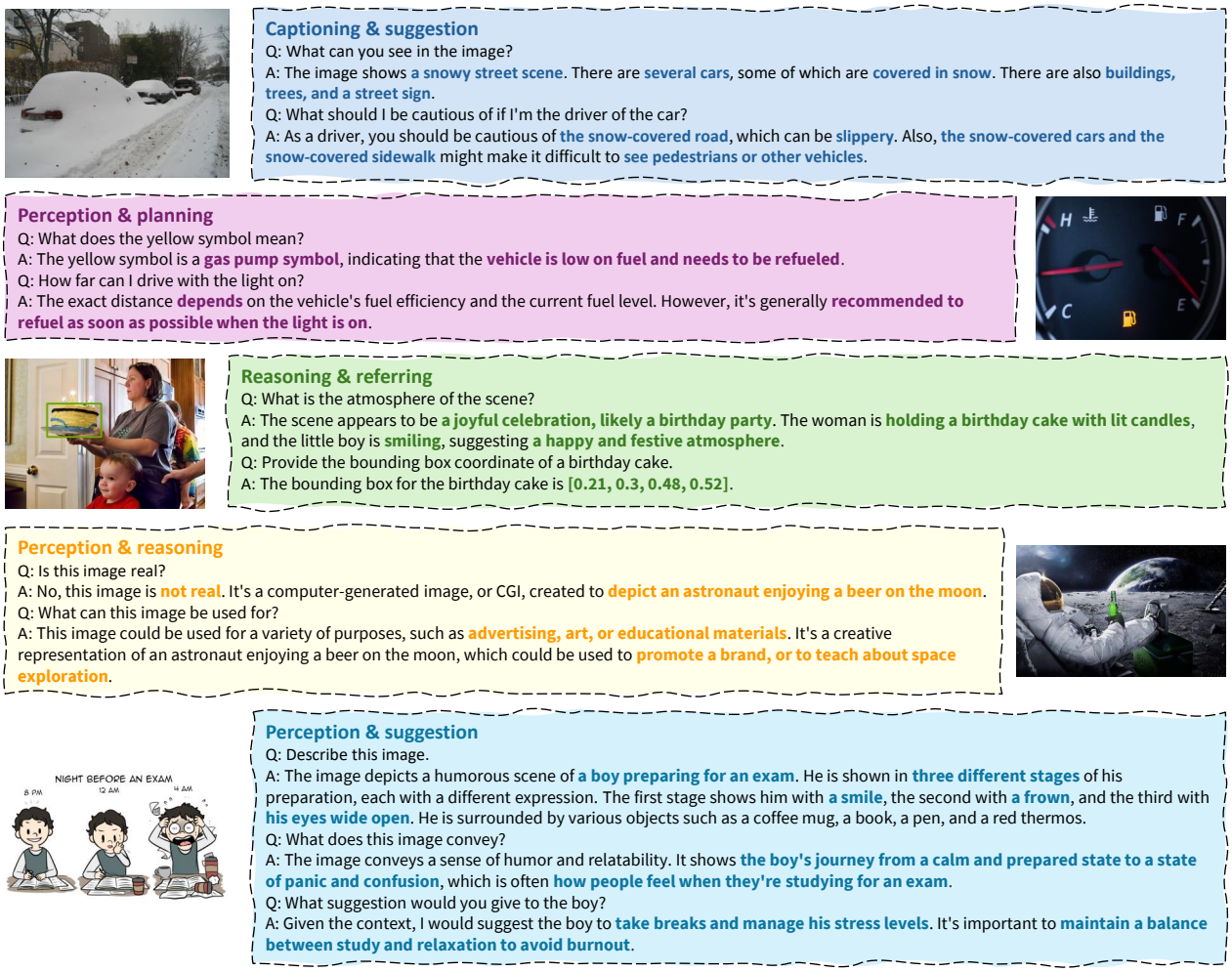

Figure 6. Demonstration of different abilities of SVIT-v1.5.

questions with "Yes" answers after random sampling from SVIT-train, ensuring a subset with exactly 80,000 samples. It is interesting that the model trained on the equilibrium, i.e. $Y : N = 1$, very likely responds "No" for any questions. The curve indicates that $Y : N = 8$ is a good data recipe for SVIT-v1.0 model and the produced model will respond "Yes" or "No" uniformly, which is close to the prior. We denote the model tuned with this data recipe as **SVIT-80K-B**, and it achieves 7.1% improvement over the model fine-tuned on SVIT-80K on MME benchmark.

Scaling Up. To investigate whether scaling up the visual instruction tuning dataset actually helps improve the model's performance, we further conduct larger experiment - training the model with SVIT-train. In the training process, the fine-tuning schedule and other hyper-parameters remain unchanged, while the learning rate is decreased from 2e-5 to 2e-6 to better fit the larger training data scale. The training takes around 24 hours on 8 NVIDIA Tesla A100 GPUs, each with 40GB memory, with DeepSpeed ZeRO Stage 3.

The evaluation results are shown in the last columns of Table 3. We compare SVIT-train to SVIT-80K (randomly selected) without any data recipe. Compared with the model fine-tuned on SVIT-80K, the total score of the model fine-tuned on SVIT-train achieves +12.7% score improvement on MME benchmark. Particularly, fine-tuning the model on more data significantly enhances the model's ability to comprehend the existence of objects (+105.6%), the color of objects (+100.0%), OCR (+39.1%), etc. The results validate the effectiveness of scaling up the visual instruction tuning dataset when fine-tuning MLLMs.

5.3. Qualitative Evaluation

In Figure 6, we provide the qualitative evaluation of the SVIT-v1.5. The first case demonstrates a conversation discussing the scene and asking for suggestions. When describing the scene, SVIT-v1.5 depicts the foreground, as well as details in the background. When giving suggestions, SVIT-v1.5 offers a comprehensive assessment, taking multiple factors into consideration, such as the snow-covered road, cars and sidewalks. In the second case, to evaluate the ability of planning, we ask the model what is happening in the image and prompt the model to plan the subsequent steps. SVIT-v1.5 accurately points out the meaning of the symbol and logically make recommendations. In terms of the ability to locate and refer objects, the third case shows that SVIT-v1.5 correctly identifies the location of the mentioned object with a bounding box, in the format of [x1, y1, x2, y2], where [x1, y1] are the normalized coordinates of the top-left point and [x2, y2] are the normalized coordinates of the bottom-right point. Regarding perception and reasoning performance, in the fourth case, SVIT-v1.5 is able to distinguish between a real image and a synthetic one. It also understands the intended use of the image, such as advertising, art or education. Similarly, the fifth case feeds SVIT-v1.5 with comics about exam preparation. SVIT-v1.5 figures out that the three sub-figures are the different stages of the preparation and infers the theme of the comics. It also generates appropriate suggestions for the character.

6. Conclusion

In this paper, we scale up visual instruction tuning by presenting a large-scale dataset – SVIT that contains in total 4.2 million instruction tuning data. We also propose new data recipe of sample selection for better diversity and balance. The abundant experiments verify that our SVIT-v1.5 trained on the proposed dataset and its subsets outperforms state-of-the-art MLLMs on multiple benchmarks.

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Appendix

A. Prompts

Based on the captions, object bounding boxes and region descriptions of images, we design four tasks and prompt GPT-4 to respond accordingly. We do not include the bounding boxes of region descriptions in the input data for *conversation*, *complex reasoning* and *detail description*, since the context length may exceed the limit of GPT-4 in many cases. The prompts share the same paragraph describing the input data at the beginning and then differ in task description, which are summarized in Figure 7. For *referring QAs*, since the location information plays a vital role in understanding the image accurately, we include the bounding boxes of region descriptions in the input data, and shorten the response to 10 QAs for every image to fit in the context limit. The prompt is summarized in Figure 8.

B. Instructions for Detail Description

Figure 9 shows the instructions for detail description. We prompt GPT-4 to generate different ways of saying "can you describe the image in detail" and accumulate all the instructions. For each image, we randomly sample one from the list as instruction.

C. Concept Set

We design the concept set with the key words for measuring the informativeness of training samples, which is illustrated in Table 4. The key words for each concept are generated by prompting GPT-4 and filtering based on their frequencies occurring in the dataset.

Table 4. The concept set and its key words for measuring the informativeness of training samples.

Concept	Key Words
color	beige, black, brown, color, gold, gray, green, khaki, lavender, mauve, olive, peach, pink, red, rose, salmon, white
material	canvas, cardboard, ceramic, cork, denim, fabric, fiberglass, foam, glass, glassy, granite, iron, latex, leather, linen, marble, mesh, metal, nylon, plaster, plastic, polymer, porcelain, satiny, silk, steel, stone, stony, suede, velvet, vinyl, wood, wooden
quantity	account, being, existence, five, four, number, one, seven, six, substance, ten, three, total, two
spatial relation	above, adjacent, ahead, backward, below, between, central, close, down, downward, far, in back, inside, left, left direction, near, on, outside, peripheral, position, proximate, remote, surrounding, under, up, upstairs, upward, without
size	big, compact, compactness, dimension, diminutive, enormity, enormous, giant, gigantic, immense, immensity, large, largeness, magnitude, massive, medium size, microscopic, miniature, minuscule, moderately, oversized, proportion, sizeable, slightly, small, smaller, vast, vastness

You are an AI visual assistant that can analyze a single image. The information of the image is made up of three parts:
(1) "captions": If it is not empty, it contains five sentences, describing the image you are observing.
(2) "objects": It contains multiple lines, each describing an object of the same image you are observing. Every line is made up of an object name and its bounding box. The bounding box is in the form of [x1, y1, x2, y2]. The values are float numbers normalized from 0 to 1, corresponding to the top left x, top left y, bottom right x, and bottom right y.
(3) "regions". It contains multiple lines, each describing a region of the same image you are observing.

Conversation:
Design three conversations between you and a person asking about this image. A conversation should include five question and answer pairs. The content within the conversation should be logically connected. You can think about the topic first and then generate the conversation according to the topic. The topic can be the visual content of the image (including the object types, counting the objects, object actions, object locations, relative positions between objects, etc.), the background knowledge of the objects, the events happening in the image, event planning, etc. In the conversation, you are called "GPT". The person talking with you is called "User".

Ask diverse questions and give corresponding answers. Only include questions that have definite answers. The answer should be in a tone that a visual AI assistant is seeing the image and answering the question. The length of the answer would better be within 50 tokens.

When using the information from the description, do not mention that the information source is the description. When using the information from the object bounding box, do not mention that the information comes from the bounding box as well. Always answer as if you are directly looking at the image.

Complex reasoning:
Create 15 plausible question and answer pairs about the image with provided information.

The question requires commonsense knowledge about the scene and can only be answered with image provided. Avoid asking questions that can be answered with commonsense knowledge alone. Avoid proposing questions that can be answered with simple visual understanding like asking about object type and color. Do not give too many details about the visual content of the image, so one has to figure it out first to answer the question correctly. The question can be asking why things happen that way, suggestions to the people in the image, etc. When providing the answer for complex questions, think step by step and include reasoning details.

When using the information from the description, do not mention that the information source is the description. When using the information from the object bounding box, do not mention that the information comes from the bounding box as well. Always answer as if you are directly looking at the image.

Desired format:
Question: ...
Answer: ...
Question: ...
Answer: ...

Detail description:
The task is describing the image in detail. Though you do not receive the pixel data of the image, utilize above textual information to think about the image and describe as if you are directly looking at the image. The description can include what people or objects are doing, object appearance, object count, object position, background details, etc. Only describe the things that you are sure about.

When using the information from the description, do not mention that the information source is the description. When using the information from the object bounding box, do not mention that the information comes from the bounding box as well.

Figure 7. The prompts of conversation, complex reasoning and detail description to GPT-4.

Referring QAs:
You are an AI visual assistant that can analyze a single image. The information of the image is made up of three parts: (1) "captions": If it is not empty, it contains five sentences, describing the image you are observing.
(2) "objects": It contains multiple lines, each describing an object of the same image you are observing. Every line is made up of the object name and its bounding box.
(3) "regions": It contains multiple lines, each describing a region of the same image you are observing. Every line is made up of the region description and the region's bounding box.

The bounding box is in the form of [x1, y1, x2, y2]. The values are float numbers normalized from 0 to 1, corresponding to the top left x, top left y, bottom right x, and bottom right y. Note that the same object may be described multiple times in "objects" with bounding boxes that are slightly different.

The task is creating 10 question and answer pairs of specific regions in the image with the provided information. The question can only be answered with image information provided. Figure out the relative positions of objects and create some questions about that. Also propose some questions that need reasoning, like why things happen that way, suggestions to the people in the image, etc. When providing answers for complex questions, think step by step and include reasoning details.

When referring to any object in the question or answer, always wrap it with prefix "<st>", suffix "<ed>" and attach its bounding box after it, in the format of "<st>man<ed> [x1, y1, x2, y2]". If multiple objects are referred to, attach all the corresponding bounding boxes after them, e.g., "<st>men<ed> [x1, y1, x2, y2], [x1, y1, x2, y2]". Refer to at least one object in the question and answer.

When using the information from the description, do not mention that the information source is the description. When using the information from the bounding box, do not mention that the information comes from the bounding box as well. Always answer as if you are directly looking at the image.

Desired format:
Question: ...
Answer: ...
Question: ...
Answer: ...

Figure 8. The prompt of referring QAs to GPT-4.

Instructions for detail description

- Can you provide a comprehensive description of the image?
- Elaborate on the details of the image.
- What are the specifics visible in the image?
- Could you offer an in-depth analysis of the image?
- Can you depict the image with precise detail?
- Give a detailed account of the image.
- Explain the image in meticulous detail.
- Can you portray the image in words?
- Give a thorough narrative of the image.
- Please provide an intricate breakdown of the image.
- Offer a complete interpretation of the image.
- Delve into the particulars of the image.
- Explain all the nuances you observe in the image.
- Provide a detailed commentary on the image.
- Illustrate the image in depth using your words.
- Could you give a blow-by-blow description of the image?
- Go into detail about the different elements of the image.
- Can you dissect the image and describe each element in detail?
- Detail the contents of the image extensively.
- Can you provide an in-depth explanation of the image?
- Provide a comprehensive overview of the image.
- Break down the elements of the image in detail.
- Can you expound upon the features of the image?
- Offer an exhaustive description of the image.
- How would you illustrate the image in words?
- Please convey the image's details verbally.
- Can you detail the contents of the image?
- Narrate what you see in the image in depth.
- Kindly provide a meticulous commentary on the image.
- Share an extensive description of the image.
- Could you interpret the image in a detailed manner?
- Present a detailed report of the image's features.
- Can you provide an intricate depiction of the image?
- Disclose every detail you see in the image.

Figure 9. Instructions for detail description.